

RP1. Adoption and use of digital technologies

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Understanding	🧠 Crystal Clear
URL	https://canvas.kth.se/courses/62723/assignments/370780
Teacher(s)	Sanna Kuoppamäki
Format	🎥 Digital
Course Offering	CM2010 VT26-4 Sociocultural Perspectives to Innovative Technologies 7.5 credits
Parent item	CM2010 Lecture 1

The purpose of the short reaction paper is to summarise, analyse and reflect the content of the lecture in relation to own field of study, user study or interests. Short reaction paper consists of 1) introducing and summarising the main content of the lecture, and presenting and explaining key concepts 2) discussing the ideas and arguments presenting in the lecture and 3) connecting the content of the lecture to your own field of study, user study or interests. The length of the short reaction paper is 1-2 pages (500-1000 words).

Reaction Paper: Socio-cultural perspectives to innovative technologies

On Tuesday, March, 17, the first lecture in the course CM2010, held by Sanna Kuoppamäki, introduced the course structure and a broader socio-cultural perspective on innovative technologies. It began with practical information about the course syllabus, learning outcomes, individual assignments, and the final group assignment on user studies for innovative health technology. It then moved into the main academic theme: how technology should be understood and how its development is shaped. I think this first lecture was a great start to the course because several central topics were briefly introduced and the overall course scope was clearly defined. I am especially looking forward to learning more about technology development in its early stages, since I believe this is one of the most difficult parts of innovation. Once a product idea has been established and its potential seems clear, it may appear easier to move toward a final product. However, the early stages require more reflection, uncertainty management, and critical decisions about users, context, and purpose.

A central message of the lecture was that technology should not only be understood through technological logic (i.e., technological determinism), but also through socio-cultural perspectives. Technical societal, cultural economic, political, and emotional factors all shape technology, just as technology shapes them in return. This shifts the focus from technology as a purely technical object to a social and cultural phenomenon that is influenced by users and contexts. Additionally, the lecture emphasized that adoption is not a linear process. Users may interpret, adapt, misuse, or integrate a product into their lives in unexpected ways. I found this especially interesting because, as a future professional in medical technology, I want to design products that take into account a wide range of factors both within and beyond the technical design itself. Therefore, considering such variables is essential if one aims to create products that genuinely affects users on multiple levels.

Another key concept introduced in the lecture was domestication of technology, which describes how a technology gradually becomes part of everyday life, how users construct experiences around it, and how they may change its purpose and meaning over time. I find this concept useful because it shows that adaptation is not defined by a single moment, but instead develops through an ongoing process consisting of four stages: appropriation, objectification, incorporation, and conversion. I agree with the idea that technology needs time to become personally

meaningful to a user. At the same time, I think that it is worth pointing out that domestication is primarily a descriptive concept, as it explains how technology evolves in relation to its users rather than functioning as a direct guide for developers. Even so, I believe the concept is still highly relevant during the design phase, because understanding how a product might eventually fit into people's lives is closely connected to the original vision behind its development, which, in turn, is guiding its potential.

In engineering education, including my own studies in medical technology, it is easy to focus mainly on technical performance and assume a "better" technology will naturally succeed. However, this lecture demonstrated that technical quality alone is not enough. A technology may work well in theory or in a laboratory, yet still fail in practise if it does not fit the realistic needs, habits, or social environments of its users. For that reason, I am convinced that defining and understanding the user is a fundamental step in product design. This is something I will reflect on more deeply in relation to my own ambition to developing meaningful and impactful products.

In summary, the lecture felt highly relevant and appreciated its content. It made clear that even the most advanced medical device or digital health system has limited value if patients, caregivers, or healthcare practitioners do not see trust it, understand it, or see how it can be integrated into their daily routines. In addition, as healthcare technologies are rarely used in solation — for instance, a person with diabetes may need to use several aids simultaneously (e.g., an insulin pen and glucose sensor) for treatment to work effectively — it means adoption depends not only on the technical design of a product, but also on the user and the wider environment in which the technology is used.